Course Problem Set (FMAM001)

Ying Ni

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LECTURE 1

Sec. 1.1. Measure Theory

Durett: Exercies 1.1.1–1.1.5;

Extra Exercise 1 (RET-Expanding a compressed proof)**.** *Read the proof of Lemma 1.1.10(i) in Durrett. The argument is very concise and several steps are implicit.*

Write out a complete and fully justified proof of the statement:

If $A, B_i \in \bar{S}$ *with*

$$A = +_{i=1}^{n} B_i,$$

then

$$\bar{\mu}(A) = \sum_{i=1}^{n} \bar{\mu}(B_i).$$

In your solution, make explicit all steps that are only sketched in the text, including the representation of sets in $\bar{S}$*, the preservation of disjointness, and the use of the definition of* $\bar{\mu}$*.*

Extra Exercise 2 (RET-Optional and recommended for students with analysis background)**.** *A proof of Carathéodory's Extension Theorem is given in the Appendix A-1. Study the proof and answer the following questions.*

1. ***What problem does the theorem solve?*** *Explain in your own words why one starts with a set function defined on an algebra, and what difficulty arises when trying to extend it to a measure on a larger* σ*-algebra.*

2. ***What is the main idea of the construction?*** *The proof first defines an outer measure* μ^* *and then restricts it to a suitable class of sets. Explain why this two-step strategy is natural:*

   $$\text{pre-measure on an algebra} \longrightarrow \text{outer measure on all subsets} \longrightarrow \text{measure on a } \sigma\text{-algebra.}$$

3. ***Why is the Carathéodory condition the right notion of measurability?*** *A set* A *is called* μ^**-measurable if*

   $$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c) \qquad \text{for all } E.$$

   Explain what this condition means conceptually, and why it is essential for turning the outer measure into a genuine measure on the class of measurable sets.

Sec. 1.2. Distributions

Durett: Exercies 1.2.1–1.2.7;

Sec. 1.3. Random variables

Durett: Exercies 1.3.1–1.3.3; 1.3.4 part i(part ii is optional); 1.3.7; solve one of Exercises 1.3.8 and 1.3.9.

LECTURE 2

Sec. 1.4. Integration

Durett: Exercises 1.4.1, 1.4.2, 1.4.4;

Sec. 1.5. Properties of integrals

Durett: Exercises 1.5.1, 1.5.2, 1.5.4, 1.5.3, 1.5.5, 1.5.6, 1.5.7, 1.5.10

Extra Exercise 3 (Reverse the Logical Order: MCT and Fatou)**.** *In Durett, Fatou's Lemma is proved first and the Monotone Convergence Theorem (MCT) is deduced from it. In this exercise, reverse the logical order.*

(a) Prove the Monotone Convergence Theorem directly:
If $0 \le f_n \uparrow f$ pointwise, then

$$\int f_n \, d\mu \uparrow \int f \, d\mu.$$

(You may first prove the result for increasing sequences of simple functions and then use approximation.)

(b) Using the Monotone Convergence Theorem, prove Fatou's Lemma:
If $f_n \ge 0$, then

$$\int \liminf_{n\to\infty} f_n \, d\mu \le \liminf_{n\to\infty} \int f_n \, d\mu.$$

(Hint: Define $g_k := \inf_{n \ge k} f_n$ and apply MCT to the increasing sequence $g_k \uparrow \liminf f_n$.)

Extra Exercise 4 (Failure of convergence theorems)**.** *Let $(Y_n)_{n\ge1}$ be random variables defined by*

$$\mathbb{P}(Y_n = y) = \begin{cases} 1 - \frac{1}{n}, & y = 0, \\ \frac{1}{n}, & y = n^2, \\ 0, & \textit{otherwise.} \end{cases}$$

1. *Compute $\mathbb{E}(Y_n)$ and show that*
$$\mathbb{E}(Y_n) \to \infty.$$

2. *Investigate whether the following theorems can be applied to justify interchanging limit and expectation:*

   - *Monotone Convergence Theorem*
   - *Dominated Convergence Theorem*
   - *Fatou's Lemma*

   For each theorem:

   - *check whether its assumptions are satisfied,*

- *if not, identify precisely which condition fails.*

3. *Show that for every $M > 0$,*

$$\lim_{n\to\infty} \mathbb{E}\big[\, Y_n \mathbf{1}_{\{Y_n > M\}} \,\big] = \infty.$$

   Interpret this result

Hint: When $n^2 > M$, the event $\{Y_n > M\}$ coincides with $\{Y_n = n^2\}$.

LECTURE 3

Sec. 1.6. Expected Value

Durett: 1.6.1, 1.6.6, 1.6.11, 1.6.12, 1.6.13, 1.6.15, 1.6.16

Extra Exercise 5 (RET)**.** *Read the proof of Theorem 1.6.8 in Durrett. The argument is concise and several steps are implicit.*

Write out a complete and fully justified proof, making explicit all steps in the truncation argument, the use of bounded convergence, Fatou's lemma, and the final estimate.

Sec. 1.7. Product Measures, Fubini's Theorem

Durett: Ex 1.7.2

Extra Exercise 6 (Expectation of a series via Fubini/Tonelli)**.** *Let $(X_n)_{n\geq 0}$ be nonnegative random variables. In Section 1.6, we showed, using MCT, that*

$$\mathbb{E}\Big(\sum_{n=0}^{\infty} X_n\Big) = \sum_{n=0}^{\infty} \mathbb{E}[X_n].$$

Show that this identity can also be obtained by viewing the sum as an integral with respect to the counting measure on $\mathbb{N}$ and applying Fubini's (more precisely Tonelli's for nonnegative functions) theorem.

Extra Exercise 7. *Read Durett Ex 1.7.1 and compare to the result in the previous exercise.*

Extra Exercise 8 (RET-Lemma 1.7.4 via Dynkin's π–λ Theorem)**.** *Read the proof of Lemma 1.7.4 in Durrett. Write out a complete and fully justified proof, making explicit all steps.*

Dynkin's π–λ ***Theorem.*** *Let Ω be a sample space. Suppose $\mathcal{C}$ is a π-system and $\mathcal{D}$ is a λ-system on Ω. If $\mathcal{C} \subset \mathcal{D}$, then $\mathcal{D} \supset \sigma(\mathcal{C})$.*

LECTURE 4

Ch 2-Independence

Durett: Ex 2.1.1 (full proof is optional, but you need to know this result), 2.1.2, 2.1.6, 2.1.10, 2.1.11, 2.1.12 (part i).

Extra Exercise 9 (RET- Elaborate Durett's proof of Theorem 2.1.9 using Theorem 2.1.7)**.** *Let $(\Omega, \mathcal{F}, P)$ be a probability space. Suppose the σ-fields $\mathcal{F}_{i,j}$, $1 \le i \le n$, $1 \le j \le m(i)$ are independent, and define*

$$\mathcal{G}_i \;=\; \sigma\left(\bigcup_{j=1}^{m(i)} \mathcal{F}_{i,j}\right).$$

Use Theorem 2.1.7 as a black box to prove that $\mathcal{G}_1, \ldots, \mathcal{G}_n$ are independent.

In your proof, make explicit:

1. *the π-system $\mathcal{A}_i$ you build inside each block i,*
2. *why $\mathcal{A}_i$ is a π-system,*
3. *why $\sigma(\mathcal{A}_i) = \mathcal{G}_i$,*
4. *and why independence holds on $\mathcal{A}_1, \ldots, \mathcal{A}_n$ (so that Theorem 2.1.7 applies).*

Ch2 (preparation)- a.s convergence v.s. convergence in probability

Extra Exercise 10 (Summability and Almost Sure Convergence)**.** *Let $Y \sim \mathrm{Exp}(1)$ and define*

$$Y_n := \frac{Y}{n}, \qquad n \ge 1.$$

1. *Prove that for every $\varepsilon > 0$,*
$$\sum_{n=1}^{\infty} \mathbb{P}(Y_n > \varepsilon) < \infty.$$
2. *Show that for every $\varepsilon > 0$, the event*
$$\{Y_n > \varepsilon \text{ infinitely often}\}$$
has probability zero.
3. *Deduce that*
$$Y_n \xrightarrow{a.s.} 0.$$
Hint: Apply part (2) to $\varepsilon_k = 1/k$ and use that $Y_n \ge 0$.

Extra Exercise 11 (Rare large values and convergence modes)**.** *Let $(Y_n)_{n\ge1}$ be random variables defined by the probability mass function*

$$\mathbb{P}(Y_n = y) = \begin{cases} 1 - \frac{1}{n}, & y = 0, \\ \frac{1}{n}, & y = n^2, \\ 0, & \textit{otherwise.} \end{cases}$$

1. *Show that*
$$Y_n \xrightarrow{P} 0.$$

2. *Determine whether Y_n converges to 0 almost surely.*

3. *For $p \geq 1$, determine whether*
$$Y_n \xrightarrow{L^p} 0.$$

Hint: Realize $Y_n = n^2 \mathbf{1}_{A_n}$ on a common probability space, where (A_n) are independent and $\mathbb{P}(A_n) = 1/n$. To test almost sure convergence, study the event that only finitely many A_n occur.

Ch2- Weak Law of Large Numbers

Durett: Ex 2.2.1; 2.2.3; 2.2.4; 2.2.6

Extra Exercise 12 (RET-Triangular Array WLLN)**.** *In this exercise you will reconstruct and explain the weak law of large numbers for triangular arrays (Theorem 2.2.11 in Durett).*

Let $\{X_{n,k} : 1 \leq k \leq n,\, n \geq 1\}$ be a triangular array of random variables such that for each n, the variables $X_{n,1}, \ldots, X_{n,n}$ are independent. Let $S_n = \sum_{k=1}^{n} X_{n,k}$ and let $b_n > 0$.
Assume:

(i) $\displaystyle\sum_{k=1}^{n} P(|X_{n,k}| > b_n) \to 0,$

(ii) $\displaystyle\frac{1}{b_n^2} \sum_{k=1}^{n} E\Big[X_{n,k}^2 \mathbf{1}_{\{|X_{n,k}| \leq b_n\}}\Big] \to 0.$

Prove that
$$\frac{S_n}{b_n} \xrightarrow{P} 0.$$

Your write-up must include:

1. *A precise definition of a triangular array and clarification of what independence is assumed.*

2. *A clear explanation of what the theorem means (in words, not only formulas).*

3. *A careful proof as if explaining the result to a fellow PhD student.*

Extra Exercise 13 (RET: Uniform Convergence of Bernstein Polynomials)**.** *In lecture we proved that for each fixed $p \in [0,1]$,*
$$f_n(p) = E\left[f\left(\frac{S_n}{n}\right)\right] \longrightarrow f(p),$$
where $S_n \sim \mathrm{Bin}(n,p)$.

In this exercise you will prove the stronger statement of uniform convergence*:*
$$\sup_{p \in [0,1]} |f_n(p) - f(p)| \to 0.$$

Your write-up must be complete and pedagogically clear.

1. ***Probabilistic representation.*** *Re-derive the identity*

$$f_n(p) = E\left[f\left(\frac{S_n}{n}\right)\right], \qquad S_n \sim \mathrm{Bin}(n,p).$$

2. ***Uniform continuity.*** *Explain why f is uniformly continuous on $[0,1]$, and show that for every $\varepsilon > 0$ there exists $\delta > 0$ such that*

$$|x-p| < \delta \quad \Rightarrow \quad |f(x) - f(p)| < \varepsilon$$

for all $x, p \in [0,1]$.

3. ***Variance computation.*** *Using Theorem 2.2.1, compute*

$$\mathrm{Var}\left(\frac{S_n}{n}\right).$$

4. ***Deviation bound.*** *Using Lemma 2.2.2 (Chebyshev's inequality), show that for every $\delta > 0$,*

$$P_p\left(\left|\frac{S_n}{n} - p\right| > \delta\right) \le \frac{p(1-p)}{n\delta^2}.$$

5. ***Uniform control.*** *Show that*

$$\sup_{p\in[0,1]} P_p\left(\left|\frac{S_n}{n} - p\right| > \delta\right) \le \frac{1}{4n\delta^2}.$$

6. ***Conclusion.*** *Combine the expectation decomposition used in lecture with the uniform deviation bound to prove*

$$\sup_{p\in[0,1]} |f_n(p) - f(p)| \to 0.$$

Important: *It is not sufficient to argue that $P_p(A_n^c) \to 0$ for each fixed p. You must obtain a bound that is uniform in p.*

Extra Exercise 14 (Proof of Theorem 2.2.11 (Truncation weak law))**.** *Let $X_{n,1}, \dots, X_{n,n}$ be independent, let $b_n \to \infty$, and define*

$$\bar{X}_{n,k} := X_{n,k}\mathbf{1}_{\{|X_{n,k}|\le b_n\}}, \qquad S_n := \sum_{k=1}^n X_{n,k}, \qquad \bar{S}_n := \sum_{k=1}^n \bar{X}_{n,k}, \qquad a_n := \sum_{k=1}^n E[\bar{X}_{n,k}].$$

Assume:

$$\text{(i)}\ \sum_{k=1}^n P(|X_{n,k}| > b_n) \to 0, \qquad \text{(ii)}\ b_n^{-2}\sum_{k=1}^n E[\bar{X}_{n,k}^2] \to 0.$$

Prove that $(S_n - a_n)/b_n \to 0$ in probability.

Suggested steps.

1. *Show the decomposition*

$$P\left(\left|\frac{S_n - a_n}{b_n}\right| > \varepsilon\right) \le P(S_n \ne \bar{S}_n) + P\left(\left|\frac{\bar{S}_n - a_n}{b_n}\right| > \varepsilon\right).$$

2. *Use the union bound to prove*

$$P(S_n \neq \bar{S}_n) \leq \sum_{k=1}^{n} P(|X_{n,k}| > b_n) \;\to\; 0.$$

3. *Show that*

$$\mathrm{Var}(\bar{S}_n) = \sum_{k=1}^{n} \mathrm{Var}(\bar{X}_{n,k}) \leq \sum_{k=1}^{n} E[\bar{X}_{n,k}^2].$$

4. *Deduce from (ii) that* $\mathrm{Var}(\bar{S}_n)/b_n^2 \to 0$ *and conclude*

$$\frac{\bar{S}_n - a_n}{b_n} \to 0 \quad \textit{in probability}$$

   using Chebyshev (or Theorem 2.2.6 / Lemma 2.2.2).

5. *Combine the steps to finish the proof.*

LECTURE 5

Ch 2- Borel Cantelli lemmas

Durett: 2.3.4, 2.3.5 (optional), 2.3.8, 2.3.11, 2.3.14

Hint to Ex 2.3.4: Recall that for any sequence (a_n) of real numbers, there exists a subsequence (a_{n_k}) such that

$$a_{n_k} \to \liminf_{n\to\infty} a_n.$$

Extra Exercise 15 (A Borel–Cantelli type result)**.** *Let $(E_n)_{n\geq 1}$ be a sequence of events such that*

$$\mathbb{P}(E_n) \leq 2^{-n} \quad \textit{for all } n \geq 1.$$

1. *Show that*
$$\sum_{n=1}^{\infty} \mathbb{P}(E_n) < \infty.$$
2. *Prove that*
$$\mathbb{P}\big(\limsup_{n\to\infty} E_n\big) = 0.$$
3. *Interpret this result in words.*

Extra Exercise 16. *Revisit Exercise 10. Use the First Borel–Cantelli Lemma to give a one-line proof of part (2).*

Extra Exercise 17. *Revisit Exercise 11.*
Construct $Y_n = n^2\mathbf{1}_{A_n}$ where (A_n) are independent events with

$$\mathbb{P}(A_n) = \frac{1}{n}.$$

Use the Second Borel–Cantelli Lemma to show that

$$Y_n \not\to 0 \quad \textit{almost surely.}$$

Compare this argument with the direct proof given earlier.

Ch 2- Strong law of large numbers

Durett: Ex 2.4.3; Ex 2.4.1 (optional)

Ch 3- Weak Convergence

Durett: (1) Solve Ex3.2.1, 3.2.11, 3.2.12, 3.2.13; (2) No need to solve but understand and know how to apply the results: 3.2.9, 3.2.10 (and optionally 3.2.5).

LECTURE 6

Ch 3- Weak Convergence (Review)

Construct an example illustrating one of the following results: Exercise 3.2.5, 3.2.9, or 3.2.10. Explain clearly why your example satisfies the assumptions and how the conclusion applies.

Ch 3- Characteristic functions and Central Limit Theorems

Extra Exercise 18 (Self-selected problems)**.** *Select 5 problems of your own from the these topics. These may be end-of-section exercises or RET exercises.*

For each selected problem:

- *state the problem clearly,*
- *provide a complete solution,*
- *briefly explain why you chose it (e.g., what concept it illustrates or why it is interesting).*

LECTURE 7

Ch 4- Conditional Expectation, Martingales,Convergence in L^p

Select 6 problems of your own from the these topics. These may be end-of-section exercises or RET exercises.

LECTURE 8

Applications

Give one application of a concept or result from this course, preferably related to your own area of research.

Describe the setting clearly, explain which probabilistic concept is used, and how it applies to the problem.

Appendix- What are RET exercises

Certain exercises in this problem set are labeled **Reconstruction and Explanation Task (RET)**.

These tasks require a complete reconstruction of a theorem or proof from the textbook (Durett), together with a clear pedagogical explanation.

A RET must contain:

1. A precise and self-contained statement of the result.
2. A clarification of all assumptions and definitions involved.
3. A complete proof in which all omitted technical details are supplied.
4. Explanatory remarks that make clear the strategy and structure of the argument.
5. When appropriate, a concrete example or illustration.

The goal is not to copy the textbook verbatim, but to demonstrate full conceptual understanding and the ability to present the argument clearly to peers. A successful RET should read like carefully written lecture notes.